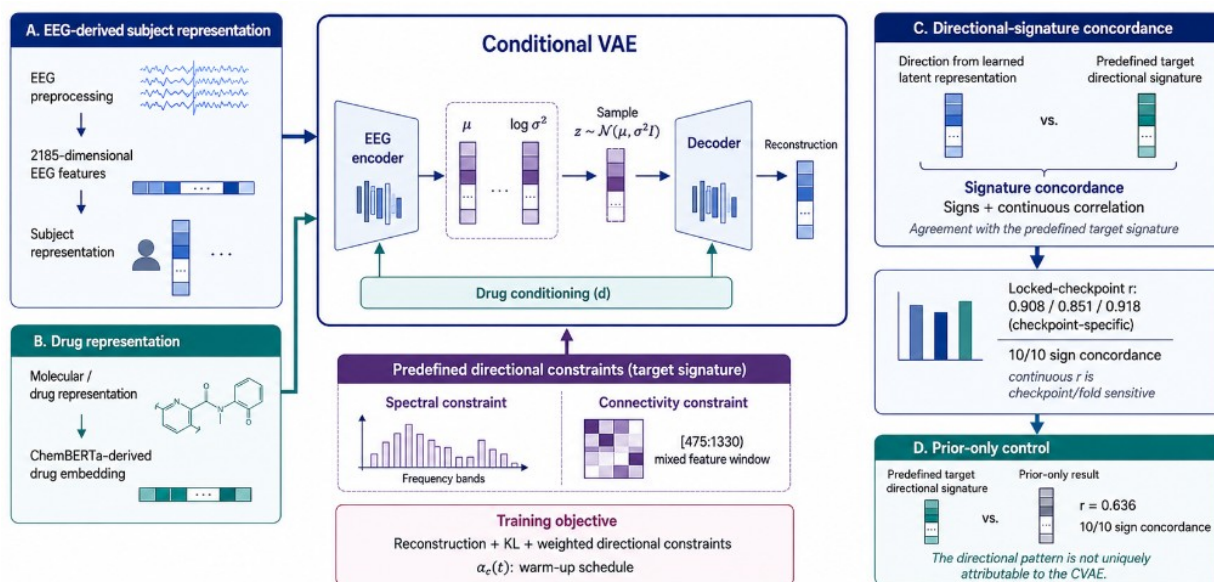


Graphical Abstract

SEME: A Decoupled Evaluation Protocol for Literature-Constrained EEG–Drug Generators

Authors to be completed at submission



Direction scoring uses the 2185-D feature-space drug-minus-baseline effect vector, not CVAE latent means. SEME treats signs, continuous r , magnitude, and encoding as separate axes.

Highlights

SEME: A Decoupled Evaluation Protocol for Literature-Constrained EEG–Drug Generators

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- SEME protocol separates signs, continuous r , magnitude, and encoding.
- Constrained signs beat unconstrained; locked high r is checkpoint-specific.
- Prior-only reaches 10/10 signs (mean $r=0.636$); circularity budget made explicit.
- No paired post-dose EEG; endpoint is signature concordance, not PD validation.
- Latent AUC 0.579 vs theta/alpha 0.675 and 2185-D 0.699; head OOF 0.593.

SEME: A Decoupled Evaluation Protocol for Literature-Constrained EEG–Drug Generators

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Abstract

Background and Objective: Chemistry-conditioned EEG generators are often scored with conflated endpoints. We introduce SEME (Signed concordance, Effect-vector fidelity, Magnitude transfer, Encoding retention), a formal evaluation protocol that scores those axes separately for literature-constrained EEG–drug generators, demonstrated on a pharmacodynamically penalized CVAE for Donepezil and Memantine.

Methods: Under SEME, a constrained CVAE ($N=66$; 2185-D EEG features; frozen ChemBERTa embeddings) used subject-level five-fold splits. External cohorts lacked paired pre/post-drug EEG, so signed concordance and continuous effect-vector r were scored against the training constrained signature on TUH, OSF, and P-ADIC. Core controls were matched unconstrained twins, a prior-only signed effect (circularity budget), linear imitation on CVAE-simulated targets, fold/seed sensitivity, and CAUEEG encoding probes against theta/alpha and packed 2185-D logistic references, with paired DeLong tests on stored out-of-fold scores.

Results: On the locked sign endpoint, constrained twins reached 10/10 agreement on TUH, OSF, and P-ADIC while matched unconstrained twins failed (3/10, 4/10, 4/10). Prior-only already reached 10/10 signs at mean $r=0.636$, so binary signs are partly prior-recoverable. Continuous r at the locked fold-0 seed-42 checkpoint was high (0.908/0.851/0.918) but checkpoint-specific (five-fold means 0.385/0.371/0.372; non-42 seeds typically 0.08 to 0.43). Magnitude did not diagnose reliably. Encoding bake-off placed latent ROC-AUC 0.579 below theta/alpha 0.675 and a 2185-D logistic reference of 0.699 (best nested-head OOF AUC 0.593; DeLong $p<0.001$ vs 2185-D).

Conclusions: SEME shows that constrained sign concordance is the more reproducible external finding in this battery, that continuous fidelity is checkpoint-specific, that prior-only bounds circularity for signs, and that residual latent diagnostic signal sits below classical feature references. The study supports a scoped measurement framework, not post-dose pharmacodynamic validation, seed-stable continuous fidelity, or standalone external diagnosis.

Keywords: SEME evaluation protocol, pharmacodynamically constrained CVAE, EEG-drug simulation, external signature assessment, encoding-level attenuation, reproducibility

1. Introduction

Chemistry-conditioned generative models are increasingly used to simulate how a named intervention would change physiological features. For such models to be scientifically usable, several endpoints must be scored separately: signed response-direction concordance with a constrained

training signature, stability of continuous effect-vector fidelity across folds and seeds, diagnostic magnitude transfer, residual diagnostic information after encoding, and incremental value beyond a literature prior. Conflating those questions invites a common failure mode in which a model looks useful on one metric while another fails. Conditional variational autoencoders (CVAEs) make that risk concrete, because reconstruction, perturbation fidelity, and downstream classification need not move together [1, 2, 3, 4]. What remains scarce is a formal evaluation design that measures those quantities on the same EEG-drug generator and localizes dissociation when they diverge.

Closest generative methods address parts of that problem. A standard CVAE conditions a Gaussian decoder without a signed pharmacodynamic penalty [1]. A β -VAE reweights KL for disentanglement rather than drug-direction priors [2]. Dr.VAE and scGen model molecular or single-cell perturbation response [3, 4]. EEG digital-twin work such as DADD targets diagnosis and prognosis rather than chemistry-conditioned response concordance [5, 6]. Alzheimer’s disease (AD) drug response is a natural stress test: acetylcholinesterase inhibitors help only a subset of patients [7, 8], and EEG slowing and connectivity changes are established AD markers with detectable cholinergic effects [9, 10, 11]. Those facts motivate constrained in silico simulation of Donepezil and Memantine. They do not license treating every latent as a clinical biomarker or every signature match as post-dose validation.

This paper contributes **SEME** (Signed concordance, Effect-vector fidelity, Magnitude transfer, Encoding retention): a formal evaluation protocol for literature-constrained, chemistry-conditioned EEG generators. The worked example is a pharmacodynamically penalized CVAE that maps 2185-dimensional resting-state EEG features and frozen ChemBERTa embeddings for Donepezil and Memantine into latent space for in silico post-drug feature simulation [12, 13]. Under SEME we score signed concordance and continuous effect-vector correlation on Temple University Hospital (TUH) EEG [14], an Open Science Framework (OSF) AD/healthy control set, and P-ADIC AD/healthy control recordings [15]; magnitude diagnosis across an escalating sequence that adds CAUEEG Normal/MCI/Dementia strata [16]; and CAUEEG encoding probes against spectral and raw-feature references. Training features use OpenNeuro ds004504 [17]. Domain shift is treated as part of the test [18].

The paper does not claim clinical validation, pharmacological efficacy, post-dose digital-twin validation, diagnostic superiority, ChemBERTa chemical generalization, or constraint-caused diagnostic compression. The contributions are:

- SEME as a decoupled protocol that treats signs, continuous r , magnitude, and encoding as separate endpoints, with matched unconstrained, prior-only (circularity budget), linear imitation, five-fold, and multi-seed controls as core method rather than afterthought.
- On that protocol, constrained twins exceed unconstrained twins on the locked sign endpoint (10/10 versus 3/10–4/10 on TUH/OSF/P-ADIC), while continuous r at locked fold-0 seed-42 (0.908/0.851/0.918) is fold- and seed-sensitive (five-fold means 0.385/0.371/0.372; non-42 seeds typically 0.08 to 0.43) and prior-only already reaches 10/10 signs at mean r = 0.636.
- Magnitude does not diagnose reliably across four escalating external tests, and a classical-versus-latent encoding bake-off retains only partial label information in μ_{base} (latent ROC-

AUC 0.579; best head OOF 0.593) below theta/alpha 0.675 and a 2185-D logistic feature-space reference of 0.699.

Section 2 defines the case-study twin and SEME endpoints. Section 3 applies the protocol. Section 4 interprets incremental value and dissociation. Section 5 concludes.

2. Methods

This section defines the case-study generator, constraints, cohorts, and the SEME endpoint families (Signed concordance, Effect-vector fidelity, Magnitude transfer, Encoding retention). Paired post-dose EEG is not part of the design. Configuration keys, scripts, and artifact paths are listed in Supplementary Material S1.

2.1. Architecture and pharmacodynamic penalties

The twin is a pharmacodynamically constrained CVAE (`constrained_mlp`): MLP encoders, no graph EEG encoder, no cross-attention, constrained loss enabled. Input EEG is a packed 2185-D vector for 19 standard 10-20 channels (PSD [0, 380), band powers [380, 475), interleaved coherence/PLV connectivity [475, 2185)). L_{band} uses delta through beta slices; L_{conn} uses window [475, 1330), which mixes early-band coherence and PLV under the interleaved flatten (S1). Layer widths and parameter count are in Table A.1. Fig. 1 summarizes data flow.

The EEG encoder maps structured tensors to a 128-D EEG latent. The drug encoder maps a frozen 384-D ChemBERTa embedding (DeepChem/ChemBERTa-77M-MLM) to a 64-D drug latent; ChemBERTa is not trained. Fusion concatenates EEG latent, drug latent, and a scalar disease label (AD = 1, HC = 0) into a 256-D representation. The CVAE encoder outputs a diagonal Gaussian in 128 dimensions with reparameterization

$$z = \mu + \exp(\tfrac{1}{2} \log \sigma^2) \odot \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I). \quad (1)$$

The decoder reconstructs the 2185-D vector. At inference, baseline uses a zero drug vector; Donepezil/Memantine use their ChemBERTa embeddings. μ_{base} denotes the zero-drug latent mean.

Signed literature priors enter as penalty terms added to reconstruction and KL [11, 9, 13, 19]:

- Donepezil: $\delta - 0.15, \theta - 0.20, \alpha + 0.25, \beta + 0.10, \text{connectivity} + 0.20$
- Memantine: $\delta - 0.10, \theta - 0.15, \alpha + 0.10, \beta + 0.05, \text{connectivity} + 0.15$

With reconstruction $L_{\text{MSE}} = \text{mean}((\hat{x} - x)^2)$, KL weight $\beta = 0.01$, band ReLU penalties L_{band} , connectivity non-decrease penalty L_{conn} (prior connectivity magnitudes are stored but not multiplied into L_{conn}), and schedule $\alpha_c(t)$ rising from 0 after a 5-epoch warmup to $\alpha_c^* = 0.1$, the total loss is

$$L = L_{\text{MSE}} + \beta L_{\text{KL}} + \alpha_c(t) L_{\text{band}} + \alpha_c(t) L_{\text{conn}}. \quad (2)$$

The KL term against $\mathcal{N}(0, I)$ is

$$L_{\text{KL}} = \frac{1}{B} \sum_{i=1}^B \left(-\frac{1}{2} \sum_{k=1}^{128} (1 + \log \sigma_{ik}^2 - \mu_{ik}^2 - \exp(\log \sigma_{ik}^2)) \right). \quad (3)$$

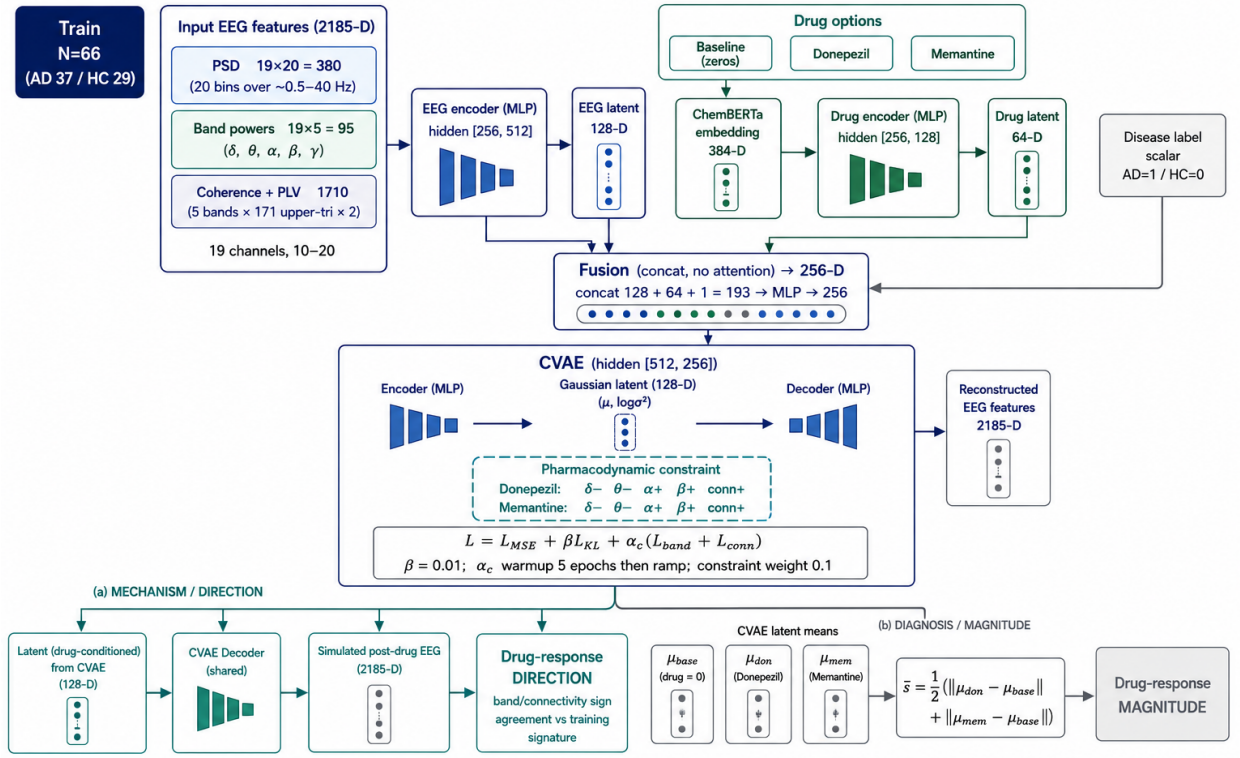


Figure 1: Architecture of the pharmacodynamically constrained conditional variational autoencoder (CVAE). Resting-state EEG features (2185-D) and ChemBERTa drug embeddings are encoded and fused with a disease label, then mapped by a CVAE whose training includes pharmacodynamic band and connectivity constraints. Training cohort: $N = 66$ (AD 37 / HC 29).

For band $b \in \{\delta, \theta, \alpha, \beta\}$ with reconstructed change Δ_b versus baseline and prior π_b ,

$$L_{band} = \sum_b |\pi_b| \text{ReLU}(-\text{sign}(\pi_b) \Delta_b), \quad (4)$$

and with connectivity-window change Δ_c ,

$$L_{conn} = \text{ReLU}(-\Delta_c). \quad (5)$$

The implemented schedule is

$$\alpha_c(t) = \begin{cases} 0 & t < T_w, \\ \alpha_c^* \min(1, (t - T_w)/5) & t \geq T_w, \end{cases} \quad (6)$$

with $T_w = 5$. Algorithm 1 summarizes training.

Input: packed EEG features $x \in \mathbb{R}^{2185}$ (PSD, band powers, interleaved coherence/PLV); ChemBERTa embedding $d \in \mathbb{R}^{384}$; disease label $y \in \{0, 1\}$; subject-level baseline features x^{base} ; priors π from `get_eeg_effect_prior`; $\beta = 0.01$; $\alpha_c^* = 0.1$; $T_w = 5$; Adam hyperparameters.

Output: checkpoint minimizing validation total loss.

Training (subject-level 5-fold; within-fold 10% validation). For each epoch $t = 0, \dots, T - 1$ and each minibatch:

1. Flatten EEG tensors to x with the same packing as the decoder target.
2. Encode EEG: $h_{\text{eeg}} = \text{EEGEncoder}(\text{PSD}, \text{band}, \text{coh}, \text{PLV})$.
3. Encode drug: $h_{\text{drug}} = \text{DrugEncoder}(d)$ (384-D ChemBERTa; enhanced 416-D prior concatenation is off for this variant).
4. Fuse: $h = \text{Fusion}(h_{\text{eeg}}, h_{\text{drug}}, y)$.
5. CVAE encode: $(\mu, \log \sigma^2) = \text{Enc}(h)$.
6. Reparameterize: $z = \mu + \exp(\frac{1}{2} \log \sigma^2) \odot \varepsilon$, $\varepsilon \sim \mathcal{N}(0, I)$.
7. Decode: $\hat{x} = \text{Dec}([z; h_{\text{drug}}; y])$ (linear output, no output activation).
8. $L_{\text{MSE}} = \text{mean}((\hat{x} - x)^2)$; L_{KL} as in Eq. (3).
9. If the batch item is not baseline, compute L_{band} and L_{conn} vs x^{base} (Eqs. (4)–(5)); else both are 0.
10. Set $\alpha_c(t)$ from Eq. (6); form L (Eq. (2)). Unique `drug_id` subsets are scored separately and summed with batch-fraction weights.
11. Backpropagate L and take one Adam step.

After each epoch, record train/val losses. Save the fold checkpoint with the lowest validation total loss.

External evaluation (not used for the Adam step): freeze the locked fold-0 checkpoint; simulate Donepezil/Memantine feature effects; score direction, magnitude, and encoding probes on held-out cohorts.

Algorithm 1: Training procedure for the pharmacodynamically constrained CVAE.

Exact index slices and packing notes are in S1.

2.2. Training cohort, freeze, and sensitivity protocol

Training features come from resting-state EEG for 66 subjects (AD 37, HC 29; OpenNeuro ds004504). Preprocessing uses FIR 0.5-40 Hz, notch 50 Hz on the training path, FastICA with empty default exclusion, 4 s epochs at 90

Splits are subject-level stratified 5-fold (`random_state=42`); within-fold validation is 10

2.3. External cohorts

Cohorts were introduced so each answered a doubt left by the previous setting (Table 1).

- **TUH** ($n = 200$; abnormal 131 / normal 69): locked features use `crop_seconds=60.0` and `apply_ica=false` in `processing_report.json`; current script default `CROP_SECONDS=320.0` was not used to regenerate locked results.
- **OSF AD/HC** ($n = 92$; HC $n = 12$).
- **P-ADIC AD/HC** ($n = 145$) [15].

Table 1: External EEG datasets used for escalating external assessment of the constrained twin. Sampling rates list native acquisition then the common 500 Hz analysis rate. Roles follow the Results narrative sequence.

Dataset	N	Groups	Labels	Sampling rate	Montage	Role
TUH	200	Abn 131 / Nor 69	Abnormal vs normal	Mixed EDF → 500 Hz	19ch 10–20	First external; proxy labels
OSF	92	AD 80 / HC 12	Folder AD vs HC	256 → 500 Hz	19ch 10–20	AD/HC; small HC
P-ADIC	145	AD 49 / HC 96	NIA-AA AD vs HC	200/500 → 500 Hz	19ch 10–20	Larger AD/HC; null check
CAUEEG	1122	Nor 436 / MCI 395 / Dem 291	Normal / MCI / Dementia	200 → 500 Hz	19ch AVG 10–20	Large-N; encoding

- **CAUEEG** Normal/MCI/Dementia (1122 features OK) for magnitude and encoding [16].

Direction tests use TUH, OSF, and P-ADIC. CAUEEG was not required for the locked 10/10 direction claim.

2.4. Evaluation endpoints and statistics

No paired pre/post-dose EEG is available. Direction measures concordance with the **training constrained signature**, a prior-shaped model-derived reference. That circularity is intentional and bounded by controls: prior-only (constant literature-signed effect; mean $r = 0.636$, 10/10), matched unconstrained CVAE ($\alpha_c = 0$), and a ridge imitation control trained on locked CVAE-simulated targets (not observed post-dose EEG).

Endpoint families (SEME). (1) **S / E:** Effect-vector Pearson r (primary continuous score) and block-level signed agreement (10/10 secondary) on cohort-mean drug-minus-baseline vectors. (2) **M:** Magnitude $\bar{s} = \frac{1}{2}(\|\mu_{\text{don}} - \mu_{\text{base}}\|_2 + \|\mu_{\text{mem}} - \mu_{\text{base}}\|_2)$. (3) **E (encoding):** CAUEEG probes on μ_{base} (disease bit fixed to 0) versus theta/alpha and packed 2185-D logistic references; nested heads use outer 5-fold / inner 3-fold search. (4) Exploratory α_c sweep on fold-0 twins. (5) Paired DeLong tests on stored subject-level OOF probe scores ($n = 727$) [20]. (6) Matched fold-0 ChemBERTa-versus-padded-one-hot ablation (same split, seed 42, $\alpha_c = 0.1$, DrugEncoder input dim 384; non-locked checkpoints only). Uncertainty for r uses subject-bootstrap percentile intervals ($B = 2000$). Magnitude uses Mann-Whitney and Cohen’s d . Probe permutation tests use $N_{\text{perm}} = 200$. No multiplicity correction was applied across the escalation sequence or DeLong pairs. Full endpoint definitions are in S1.

3. Results

Results follow the SEME axes. The primary positive contrast is constrained versus unconstrained signed concordance. Continuous high r at one locked checkpoint is reported secondarily. Prior-only quantifies the circularity budget for binary signs.

Table 2: Locked unconstrained battery versus constrained direction agreement against the constrained training signature (locked fold-0 seed-42 checkpoints). Unconstrained r values are from that locked battery, not from the α_c -sweep fold-0 re-simulation. Prior-only signed effects also reach 10/10 on signs with lower mean effect-vector correlation ($r=0.636$).

Dataset	n	Constr. agree	Constr. r	Unconstr. agree	Unconstr. r
TUH	200	10/10	0.908	3/10	-0.309
OSF	92	10/10	0.851	4/10	-0.331
P-ADIC	145	10/10	0.918	4/10	-0.327

3.1. Controls: unconstrained, prior-only, and linear imitation

Matched unconstrained twin (primary contrast). On the locked fold-0 comparison against the constrained training signature, unconstrained agreement was 3/10 on TUH ($r = -0.309$), 4/10 on OSF ($r = -0.331$), and 4/10 on P-ADIC ($r = -0.327$). Constrained agreement on the same cohorts is 10/10 (Section 3.2). Thus the CVAE-with-constraint adds signed concordance that the matched unconstrained twin does not deliver under this endpoint. Unconstrained magnitude contrasts remained null or non-diagnostic, and the unconstrained CAUEEG μ_{base} probe reached mean ROC-AUC 0.539 (below constrained 0.579 and below theta/alpha 0.675). Table 2 summarizes the locked direction contrast.

Prior-only baseline (circularity budget). A constant literature-signed 2185-D effect, without a CVAE, yielded 10/10 signs at mean $r = 0.636$ (Donepezil 0.572; Memantine 0.700). Binary signs are therefore partly recoverable from the prior alone. SEME treats this as an explicit circularity budget: the CVAE is not claimed to uniquely explain 10/10 signs.

Linear imitation control. A ridge map trained on packed baselines, frozen drug embeddings, and disease labels to predict locked CVAE-simulated drug-minus-baseline vectors recovered 10/10 signs on TUH/OSF/P-ADIC with $r = 0.851, 0.639$, and 0.775 . At the same fold-0 comparison, the locked constrained CVAE remained higher in continuous r (0.908, 0.851, 0.918). This is imitation of CVAE-generated targets, not independent biological validation.

3.2. Direction concordance and continuous- r sensitivity

Locked fold-0 seed-42 constrained correlations were $r = 0.908$ (TUH, $n = 200$), 0.851 (OSF, $n = 92$), and 0.918 (P-ADIC, $n = 145$), each with secondary signs 10/10 (Table 3; Fig. 2). Subject-bootstrap 95

Five-fold means were 0.385 ± 0.308 , 0.371 ± 0.294 , and 0.372 ± 0.310 ; signs were 10/10 on three folds, 9/10 on fold 4, and 8/10 on fold 2. Holding fold membership fixed and varying train seeds {42, 7, 21, 123, 2024}, non-42 continuous r typically ranged 0.08-0.43 while signs remained mostly 9/10-10/10 (Table 4). Matched unconstrained multi-seed runs stayed low/negative in continuous r ; constrained signs exceeded unconstrained signs in all 15 seed $\times$ cohort cells without establishing continuous- r stabilization. An equal-weight fold ensemble ($r = 0.642/0.626/0.627$) did not clearly exceed prior-only mean $r = 0.636$.

Thus under SEME the more reproducible external component in this battery is directional/sign concordance; continuous high r at locked seed-42 is not a stable architecture-level property.

Table 3: Checkpoint-specific external signature concordance at the locked fold-0 seed-42 checkpoint. Band-level direction agreement is scored between each external cohort and the training-cohort constrained signature on five feature blocks $\times$ two drugs (Donepezil, Memantine; maximum 10/10). Correlation r is the Pearson correlation of mean drug-minus-baseline effect vectors (effect_magnitude_correlation). These values are not architecture-wide performance estimates; fold and training-seed sensitivity are reported in the Results. CAUEEG is not included in the locked 10/10 direction claim.

Dataset	N	Agree	r	Assessment role
TUH	200	10/10	0.908	1st external assessment
OSF	92	10/10	0.851	2nd assessment (independent)
P-ADIC	145	10/10	0.918	3rd assessment (acq. change)

Table 4: Fixed-fold training-seed sensitivity of external direction metrics. Fold assignment and fold-0 subject membership were frozen at split_seed = 42; only train_seed varied. Seed 42 is the locked historical checkpoint (checkpoint_constrained.pt); seeds 7, 21, 123, and 2024 are independent training-seed perturbations under the same fold split. Values are Pearson r (mean Donepezil/Memantine effect-vector correlation) and sign agreement. Source: models/validation/multiseed_robustness/aggregate_results.csv.

Seed	Role	TUH (r ; agree)	OSF (r ; agree)	P-ADIC (r ; agree)
42	Locked fold-0	0.908; 10/10	0.851; 10/10	0.918; 10/10
7	Alternate train seed	0.408; 10/10	0.408; 10/10	0.428; 10/10
21	Alternate train seed	0.154; 10/10	0.160; 10/10	0.106; 8/10
123	Alternate train seed	0.147; 9/10	0.230; 9/10	0.080; 9/10
2024	Alternate train seed	0.163; 10/10	0.182; 10/10	0.146; 10/10
Mean r (all five)		0.356	0.366	0.336
Median r		0.163	0.230	0.146
SD r		0.328	0.288	0.354

3.3. Magnitude escalation

Magnitude $\bar{s}$ was tested in sequence (Table 5; Fig. 3). TUH abnormal versus normal was null ($p = 0.272$, $d = -0.13$). OSF AD versus HC was an exploratory positive with HC $n = 12$ caveats ($p = 0.000278$, $d = 0.44$). P-ADIC AD versus HC was null ($p = 0.968$, $d = -0.07$). CAUEEG Dementia versus Normal was null ($p = 0.516$, $d = 0.00$, $n = 727$), despite clear theta/alpha slowing on the same features ($p < 0.001$, $d = 0.47$) and good scale/PCA overlap (Fig. 4). Secondary CAUEEG contrasts and disease-label ablations remained null. Magnitude therefore does not provide reliable external diagnosis in this escalation.

3.4. Encoding bake-off (classical features versus latent)

On CAUEEG Dementia versus Normal ($n = 727$), SEME’s Encoding axis compares classical feature references to frozen-twin μ_{base} readouts on the same labels (Fig. 5; Fig. 6; Table 6). A logistic probe on μ_{base} (disease bit fixed to 0) reached mean ROC-AUC 0.579 (permutation $p =$

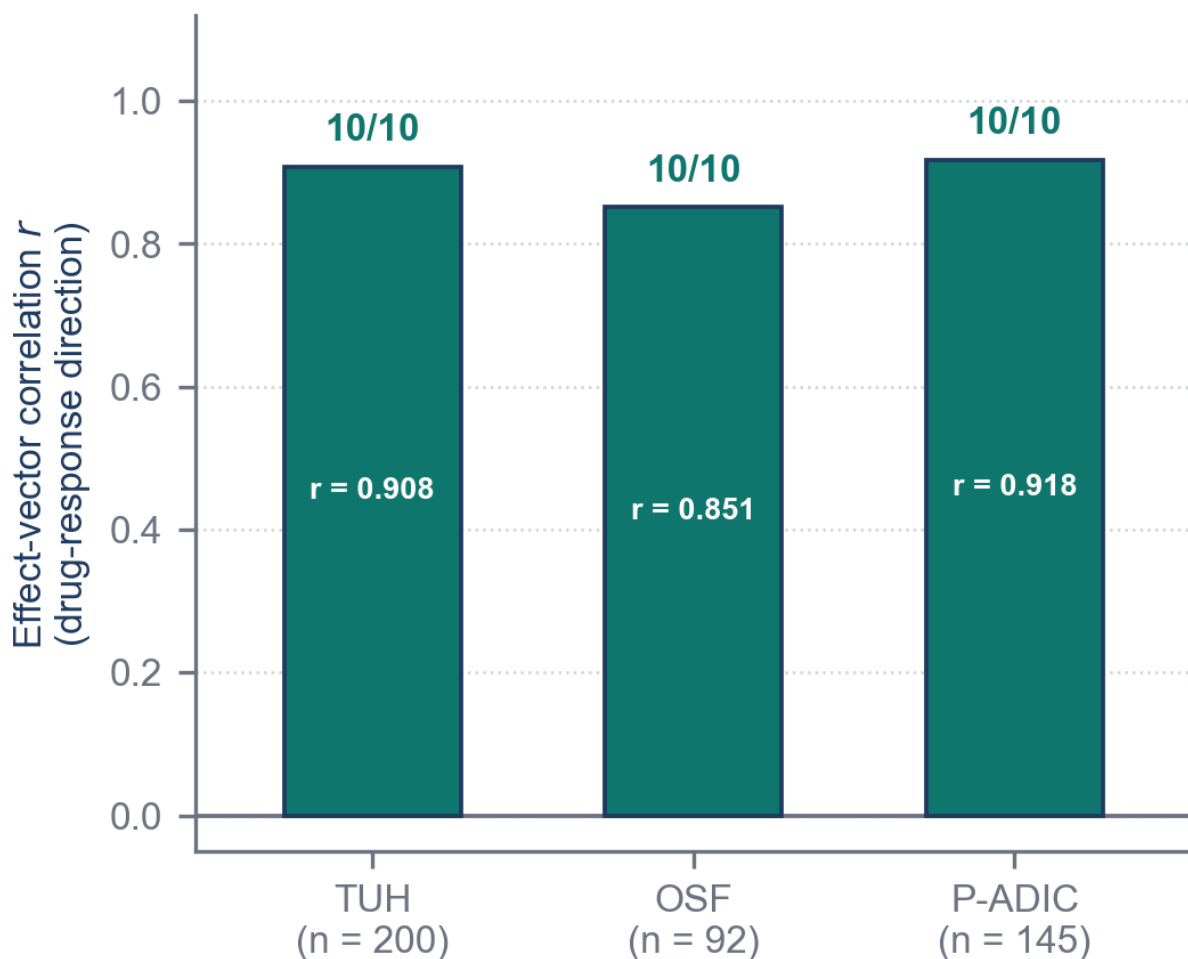


Figure 2: Locked fold-0 seed-42 checkpoint only: external signature concordance scored against the training constrained signature. Bars show full-sample effect-vector correlation r for TUH ($r=0.908$, $n=200$), OSF ($r=0.851$, $n=92$), and P-ADIC ($r=0.918$, $n=145$). These bars are not representative of five-fold or multi-seed performance. Secondary signed agreement was 10/10 on each cohort. These values are checkpoint-specific signature concordance, not independent pharmacodynamic validation.

0.010). Theta/alpha reached 0.675 (OOF 0.673) and a packed 2185-D logistic reference reached 0.699 (OOF 0.699); 0.699 is a feature-space reference, not a CVAE score or ceiling. Nested heads peaked at OOF AUC 0.593 (tuned L2 logistic; permutation $p = 0.005$); MLP and boosting did not close the gap. Paired DeLong tests on stored OOF scores confirmed latent below theta/alpha ($z = -3.63$, $p = 0.000285$) and below 2185-D ($z = -4.64$, $p < 0.001$), and best head below theta/alpha ($z = -3.32$, $p = 0.000909$) and below 2185-D ($z = -4.09$, $p < 0.001$); theta/alpha versus 2185-D was not significant ($z = -1.11$, $p = 0.268$). Matched unconstrained encoding did not exceed constrained latent AUC (0.539 vs 0.579). The bake-off shows encoding-level attenuation under tested readouts, not a formal information-bottleneck proof.

Table 5: Twin latent drug-response magnitude discrimination across escalating external cohorts. Status strings are taken from the validation JSON (FAIL, PASS_WITH_CAVEATS). The OSF primary row uses `honest_verdict.status`; OSF disease-label sensitivity rows have no status field (shown as –). The Kruskal–Wallis row reports H instead of Cohen’s d and Kruskal p in the p column.

Dataset	Contrast	N (A / B)	p	d	Status
TUH					
TUH	Abnormal vs normal	131 / 69	0.272	-0.13	FAIL
OSF					
OSF	AD vs HC	80 / 12	$p < 0.001$	0.44	PASS*
OSF	AD vs HC (disease=0)	80 / 12	$p < 0.001$	0.64	–
OSF	AD vs HC (disease=1)	80 / 12	$p < 0.001$	0.34	–
P-ADIC					
P-ADIC	AD vs HC	49 / 96	0.968	-0.07	FAIL
P-ADIC	AD vs HC (disease=0)	49 / 96	0.527	-0.16	FAIL
P-ADIC	AD vs HC (disease=1)	49 / 96	0.702	-0.14	FAIL
CAUEEG					
CAUEEG	Dementia vs Normal	291 / 436	0.516	-0.00	FAIL
CAUEEG	Dementia vs MCI	291 / 395	0.877	0.07	FAIL
CAUEEG	MCI vs Normal	395 / 436	0.363	-0.07	FAIL
CAUEEG	AD-tagged Dem. vs Normal	214 / 436	0.624	-0.01	FAIL
CAUEEG	Kruskal–Wallis (3 groups)	436 / 395 / 291	0.637	$H=0.90$	FAIL
CAUEEG	Dem. vs Normal (disease=0)	291 / 436	0.411	-0.04	FAIL
CAUEEG	Dem. vs Normal (disease=1)	291 / 436	0.776	0.00	FAIL

*PASS_WITH_CAVEATS (HC $n=12$; exploratory).

Table 6: CAUEEG Dementia vs Normal encoding analysis ($n = 727$). Out-of-fold (OOF) ROC-AUC and balanced accuracy are reported. Permutation p -values are included where computed (original latent probe: 200 label shuffles of 5-fold CV; best head: 200 shuffles of nested CV). The 2185-D logistic row is a raw feature-space diagnostic reference, not a CVAE result, not a theoretical upper bound, and not a supervised deep 2185-D classifier. The final column is OOF AUC minus the θ/α pooled OOF AUC (0.673; mean-fold θ/α AUC = 0.675 is used in the prose and Fig. 5). Complements Fig. 5.

Model	Input	AUC	Bal. acc.	Perm. p	vs. θ/α
2185-D logistic (feature ref.)	packed 2185-D	0.699	0.642	–	+0.026
θ/α probe (ref.)	θ/α	0.673	0.625	–	+0.000
Untuned logistic	μ_{base}	0.576	0.568	0.010	-0.097
Tuned L2 logistic	μ_{base}	0.593	0.573	0.005	-0.080
MLP	μ_{base}	0.543	0.543	–	-0.130
HistGradientBoosting	μ_{base}	0.556	0.548	–	-0.117
Tuned LogReg ($\mu + \log\text{var}$)	$\mu + \log\text{var}$	0.587	0.554	–	-0.085

3.5. Constraint-weight sweep

An exploratory fold-0 α_c sweep is summarized in Fig. 7. Signs rose to 10/10 by $\alpha_c = 0.05$, continuous TUH r peaked near the paper weight $\alpha_c = 0.1$ ($r = 0.897$), and latent AUC was non-monotone (0.539, 0.527, 0.579, 0.514, 0.578 at 0, 0.05, 0.1, 0.2, 0.5). The sweep does not

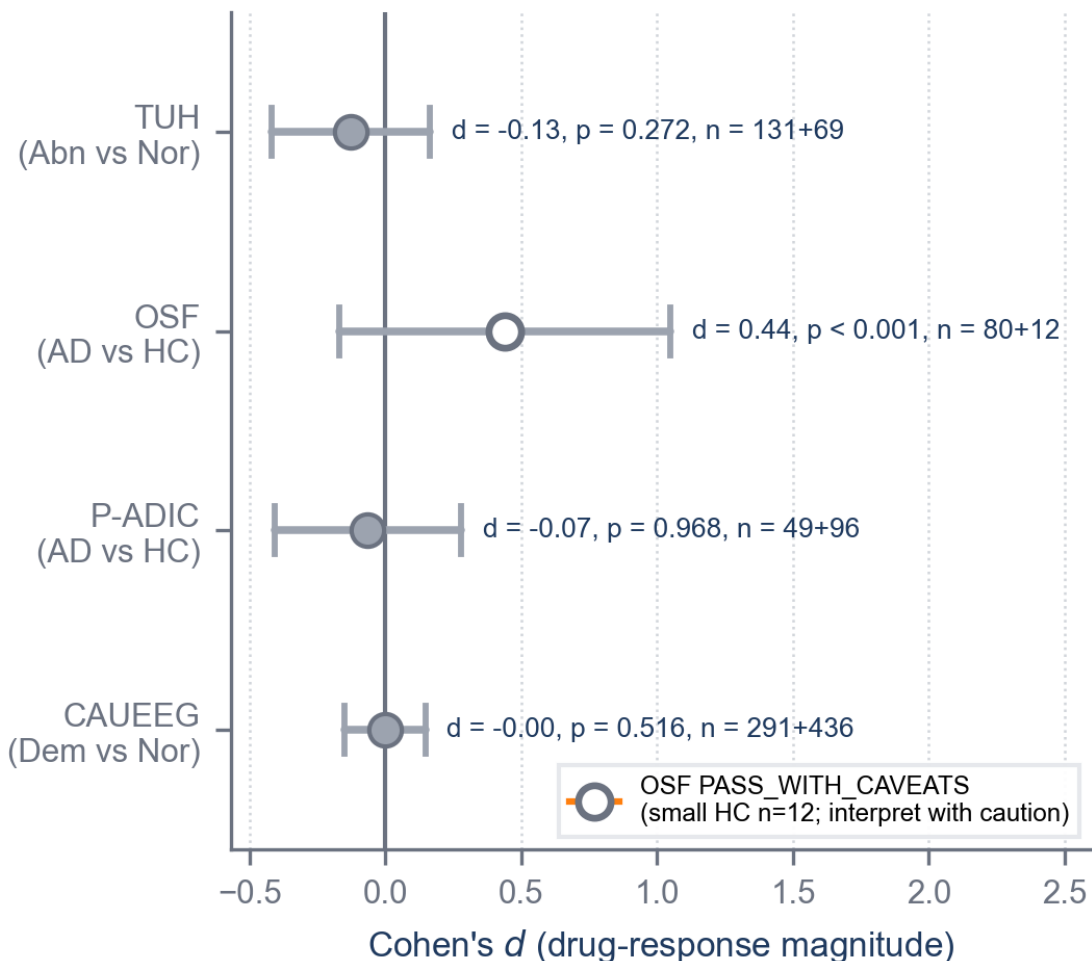


Figure 3: Forest plot of Cohen’s d for primary twin magnitude contrasts on TUH, OSF, P-ADIC, and CAUEEG. Approximate 95% intervals use the Hedges–Olkin standard-error formula from reported d and group sizes. Exact n , p , and d values are listed in Table 5.

support a simple direction-versus-diagnosis trade-off and is not a nested selection of α_c .

3.6. ChemBERTa versus padded one-hot

A matched fold-0 constrained retrain replaced ChemBERTa vectors with 384-D padded one-hot drug IDs without touching the locked checkpoint. Against the locked training signature, one-hot reached 10/10 signs on TUH, OSF, and P-ADIC ($r = 0.176/0.194/0.174$), while the rematched ChemBERTa arm reached 9/10 ($r = 0.104/0.110/0.076$). Both rematched continuous- r values sit in the low multi-seed regime, not the locked high- r checkpoint. CAUEEG latent mean AUC was 0.540 (ChemBERTa rematch) versus 0.559 (one-hot). In this two-drug setting, chemical embeddings are not required for constraint-consistent signs; the ablation is not a large-library ChemBERTa generalization test, and the locked ChemBERTa twin remains the primary historical checkpoint.

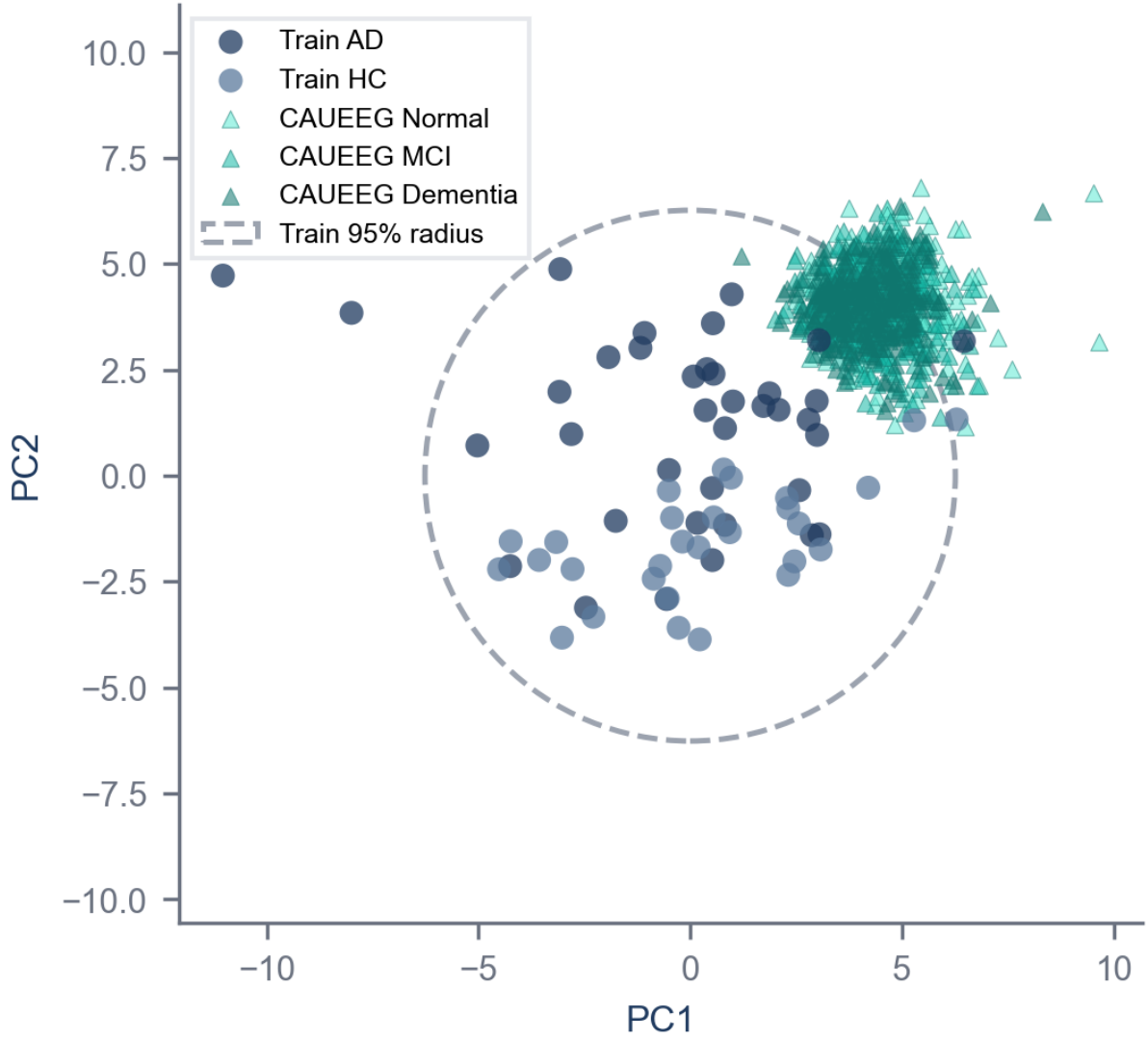


Figure 4: PCA of 2185-D features: training cohort versus CAUEEG. Dashed circle marks the training 95% radius; overlap fraction within that radius is 0.611.

4. Discussion

SEME separates four quantities that generative EEG-drug papers often conflate. In this battery, the clearest incremental finding is constrained versus unconstrained signed concordance: constrained twins reached 10/10 on TUH, OSF, and P-ADIC while matched unconstrained twins failed (3/10–4/10), and fixed-fold multi-seed runs mostly retained 9/10 to 10/10 constrained signs. Continuous external r is not an architecture-level property: locked values 0.908/0.851/0.918 fall to five-fold means near 0.37 to 0.39 and to typical non-42 values of 0.08 to 0.43. Prior-only already reaches 10/10 signs at mean $r = 0.636$, and the fold ensemble (0.642/0.626/0.627) does not clearly exceed that prior, so the CVAE is not claimed to uniquely explain binary signs or to

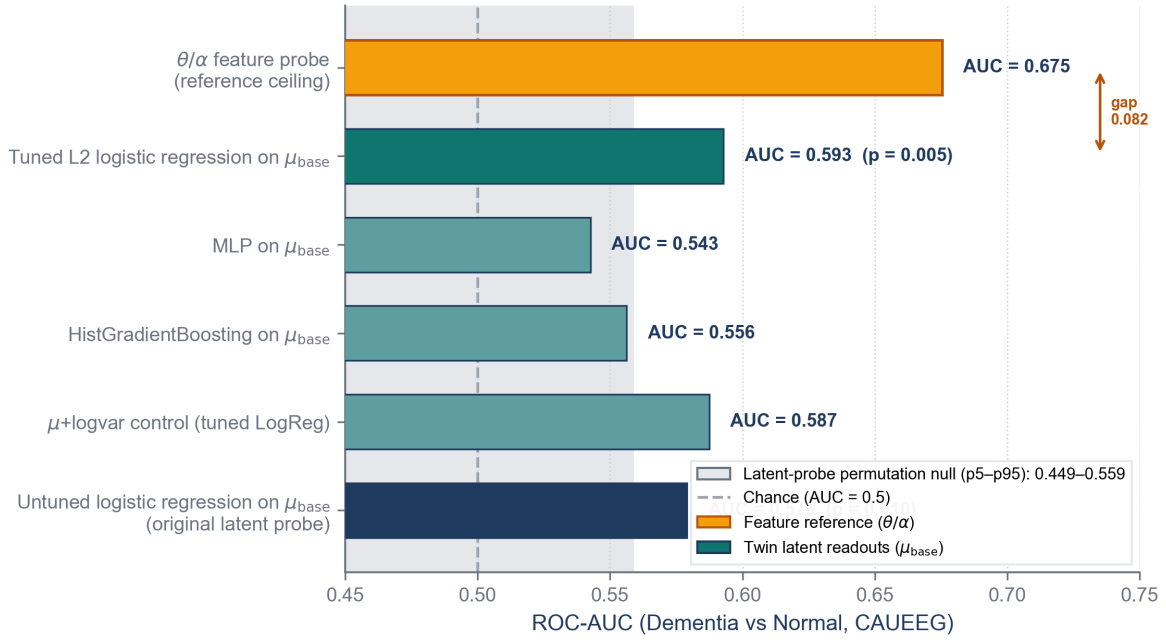


Figure 5: SEME Encoding bake-off on CAUEEG Dementia versus Normal ($n=727$). Amber: classical theta/alpha and related feature references. Teal/navy: frozen-twin μ_{base} readouts. Gray band: latent-probe permutation null 5th–95th percentiles (0.449–0.559). Best nested head OOF AUC = 0.593. A packed 2185-D logistic feature-space reference reaches mean-fold AUC = 0.699 (not a CVAE score).

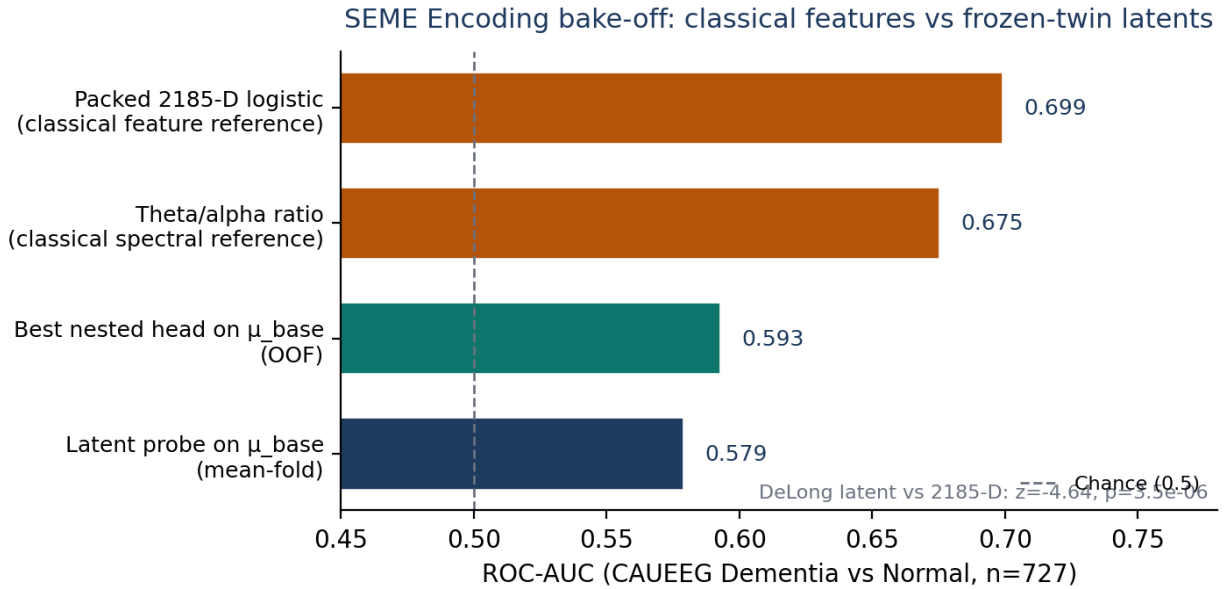


Figure 6: Compact SEME Encoding bake-off panel: classical feature references versus latent/head readouts on CAUEEG Dementia versus Normal ($n=727$). Values match locked probe/head artifacts; DeLong annotation uses the non-locked Phase C OOF comparison when available.

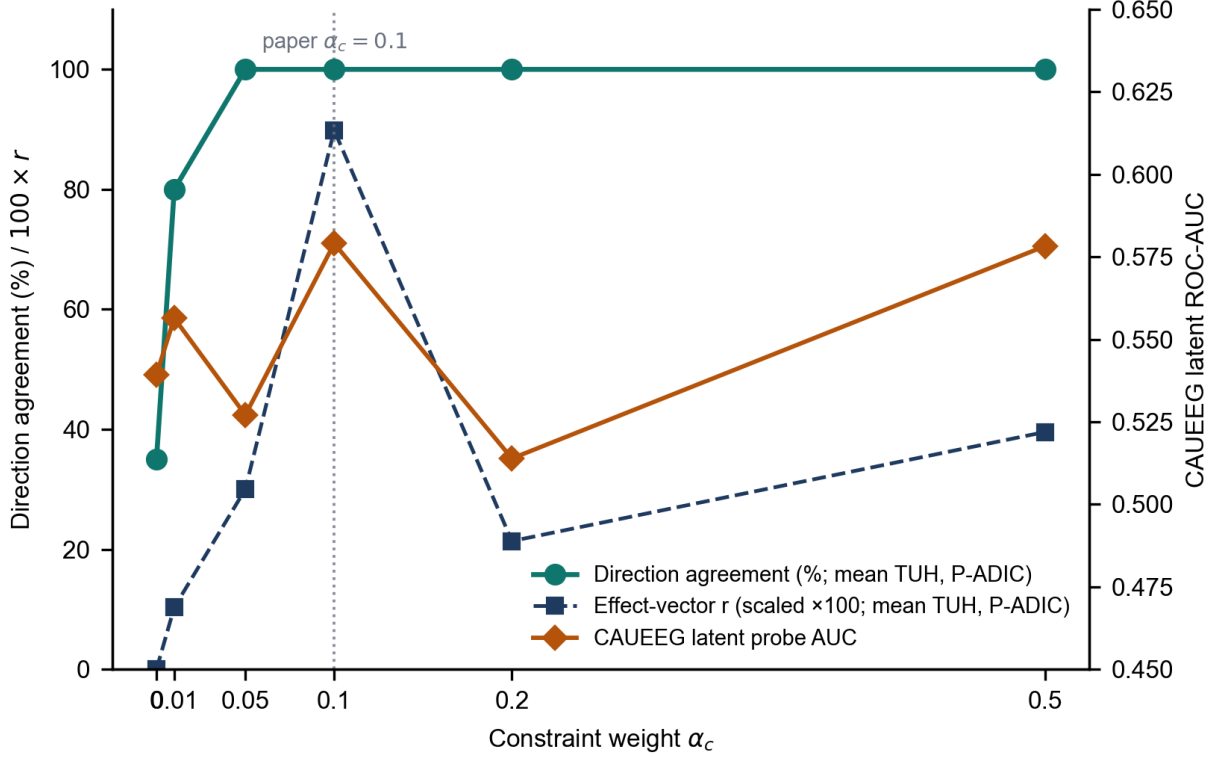


Figure 7: Exploratory fold-0 constraint-weight (α_c) sweep: signed direction agreement and continuous effect-vector correlation, with CAUEEG latent-probe AUC. Not a nested selection of α_c .

have independently discovered drug direction. Magnitude diagnosis fails across the escalation sequence, and the classical-versus-latent encoding bake-off retains only partial label information (0.579; best head 0.593) below theta/alpha 0.675 and a 2185-D logistic feature-space reference of 0.699.

What the CVAE adds under SEME (incremental-value box). (i) Relative to a matched unconstrained twin, pharmacodynamic penalties buy signed concordance with the training constrained signature on the locked external cohorts. (ii) Relative to prior-only, the locked continuous- r checkpoint can exceed the prior’s mean $r = 0.636$, but that continuous advantage is checkpoint-specific and is not established as fold- or seed-stable. (iii) Relative to classical CAUEEG feature probes, frozen μ_{base} readouts do not win the encoding bake-off. SEME’s job is to keep those three comparisons separate.

Relative to diagnosis-oriented EEG twins such as DADD [5, 6], this case study occupies a different niche: ChemBERTa-conditioned simulation under explicit band/connectivity penalties, scored as a measurement framework rather than as a clinically validated pharmacodynamic predictor [21]. The dissociation parallels known separations among reconstruction, perturbation fidelity, and predictive utility in VAEs [1, 3]. Practically, the twin is best used for constraint-consistent signature concordance and hypothesis generation. If diagnosis is required, calibrate against feature-level references rather than frozen μ_{base} .

What this study establishes. SEME as a reusable endpoint taxonomy; feasibility of a literature-constrained CVAE with reproducible implementation details; external sign concordance under the locked protocol; constrained-versus-unconstrained and prior-only/imitation controls on that endpoint; separation of signs, continuous r , magnitude, and encoding; and encoding attenuation under tested readouts relative to feature references.

What it does not establish. Empirical post-dose pharmacodynamic validation; independent pharmacological discovery; seed- or fold-stable continuous r ; diagnostic superiority; ChemBERTa generalization; an information-theoretic bottleneck; a supervised deep 2185-D classifier; superiority over prior-only on signs; or constraint-caused diagnostic compression (unconstrained latent AUC was not higher than constrained).

Limitations are specific: no paired post-dose EEG; $N = 66$ versus multilayer capacity; freeze documented but not auditable preregistration; linear imitation uses CVAE-simulated targets; empty default ICA exclusion; CAUEEG dementia-label heterogeneity; exploratory α_c sweep; rematched ChemBERTa/one-hot arms do not recover the locked high- r regime. Future work includes broader seed $\times$ fold grids under the same SEME taxonomy and a public tagged release of the regeneration package.

5. Conclusion

This paper contributes SEME (Signed concordance, Effect-vector fidelity, Magnitude transfer, Encoding retention), a formal evaluation protocol for literature-constrained, chemistry-conditioned EEG generators. On a pharmacodynamically penalized Donepezil/Memantine CVAE case study, constrained twins exceeded unconstrained twins on the locked sign endpoint (10/10 versus 3/10–4/10 on TUH/OSF/P-ADIC), while continuous fidelity at locked fold-0 seed-42 ($r = 0.908/0.851/0.918$) was fold- and seed-sensitive, prior-only already achieved 10/10 signs at mean $r = 0.636$, magnitude diagnosis did not transfer, and an encoding bake-off left latent probes (0.579; best head OOF 0.593) below feature references (0.675 and 0.699). The findings support a scoped measurement framework for constrained EEG-drug simulation, not a clinically validated pharmacodynamic digital twin, not seed-stable continuous external fidelity, and not standalone external diagnosis.

CRedit authorship contribution statement

Author contributions will be completed when the author list is finalized. No corresponding author is designated in this draft.

5.1. Ethics statement

This work is a secondary analysis of de-identified EEG datasets. No new human participants were recruited and no clinical interventions were performed. Training resting-state EEG features derive from the public OpenNeuro accession ds004504 (Miltiadous et al.). External cohorts were obtained under each provider’s academic data-use terms: the Temple University Hospital (TUH) EEG Corpus data use agreement; a public Open Science Framework (OSF) Eyes_closed AD/HC partition; the Dryad P-ADIC release (DOI 10.5061/dryad.8gtht76pw); and academic,

non-commercial access to CAUEEG granted by the CAUEEG team (access contact: Min-jae Kim), with no redistribution of raw CAUEEG or TUH recordings. Analyses were conducted for academic research. The study does not claim prospective clinical decision support.

5.2. Funding

This research received no specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Data availability

Training resting-state EEG features used in this project are from OpenNeuro accession ds004504 (Miltiadous et al. AD/HC resting EEG), processed to the feature arrays described in Methods. Temple University Hospital (TUH) clinical EEG is available to researchers under the TUH corpus data use agreement and cannot be redistributed by the authors. The OSF Eyes_closed AD/HC features used for the AD/HC external magnitude and direction checks were imported from a public OSF-hosted partition already present in the project archive. P-ADIC AD and control matrices are available from Dryad (DOI 10.5061/dryad.8gtht76pw) under the release terms of Shor et al. CAUEEG raw recordings were obtained by academic application to the CAUEEG team (access granted by Min-jae Kim; Kim et al., NeuroImage 2023) for non-commercial research use and must not be redistributed. Derived project artifacts that can be shared subject to those upstream terms include processed feature arrays generated in this repository (where redistribution is allowed), validation JSON summaries, and analysis code. Raw CAUEEG and TUH EDFs will not be redistributed by the authors. File-level paths are listed in Supplementary Material S1.

5.3. Code availability

Analysis and figure-generation code for this manuscript live in the local project repository and are not currently in a public repository. The Python environment is the project virtualenv `eeg_twin` (Python 3.10); install from `requirements.txt` into that environment. Exact script names, configuration keys, checkpoint aliases, and regeneration commands are listed in Supplementary Material S1 and `paper/CODE_AVAILABILITY.md`. A public mirror (URL and commit hash) will be linked at acceptance; until then, the packaged Overleaf zip and the local repository are the reproducibility sources for the reported JSON and figures.

Glossary

Terms below follow Methods and Results. They do not introduce new endpoints or claims.

- **CVAE**: conditional variational autoencoder.
- **EEG**: electroencephalography.
- **PD**: pharmacodynamic. In this paper the label names the literature-prior training objective, not empirical post-dose pharmacodynamic validation.

- **ROC-AUC:** area under the receiver operating characteristic curve.
- **OOF:** out-of-fold.
- **TUH:** Temple University Hospital EEG corpus.
- **OSF:** Open Science Framework Eyes_closed AD/HC set used here.
- **CAUEEG:** Chung-Ang University Hospital EEG dataset.
- **P-ADIC:** p-adic quantum potential EEG cohort (Shor et al.; Dryad).
- **ChemBERTa:** chemical Bidirectional Encoder Representations from Transformers. Here, frozen precomputed embeddings for Donepezil and Memantine, not a large-library chemistry test.
- **MSE:** mean squared error (reconstruction term).
- **KL:** Kullback-Leibler divergence (CVAE regularizer).
- **ICA:** independent component analysis (FastICA on the training path).
- **FIR:** finite impulse response filter (training band-pass 0.5-40 Hz).
- **PCA:** principal component analysis.
- **Latent representation:** CVAE latent mean under specified conditioning (baseline: zero-drug). Residual diagnostic content is scored with probes, not as an information-theoretic bottleneck.
- **Drug-response signature:** the training constrained signature (mean simulated drug-minus-baseline feature vector). External direction scoring is concordance with that signature, not post-dose pharmacodynamic validation.
- **Digital twin:** the constrained CVAE used for in silico EEG-drug signature simulation. It is not a clinically validated pharmacodynamic model.
- **Effect vector:** cohort-mean drug-minus-baseline feature vector used to compute Pearson r and block-level sign agreement.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We thank the Temple University Hospital (TUH) EEG Corpus team for access to clinical EEG under the TUH data use agreement; Min-jae Kim and the CAUEEG investigators for academic access to the Chung-Ang University Hospital EEG dataset; and the providers of OpenNeuro ds004504 (Miltiadous et al.), the OSF Eyes_closed AD/HC partition, and the Dryad P-ADIC release. We also thank the Alzheimer’s Disease Neuroimaging Initiative (ADNI) for related clinical data infrastructure used in supporting cohort work outside the primary EEG external battery.

Appendix A. Reproducibility appendix

This appendix lists configuration keys, scripts, and artifact paths used to regenerate the manuscript numbers. It is not required to follow the scientific argument in the main text. Table A.3 lists methodological safeguards and remaining scope boundaries, including non-locked Phase C DeLong and ChemBERTa-versus-one-hot rows.

Appendix A.1. Reproduction contract

- Environment: Python 3.10.0 in eeg_twin/; PyTorch 2.10.0+cu128; NumPy 1.26.4; SciPy 1.15.3; scikit-learn 1.7.2; MNE 1.11.0; Transformers 5.0.0 (computational_requirements.json).
- Hardware: NVIDIA GeForce RTX 5060 Ti (16 GB).
- Split: subject-level StratifiedKfold (n_splits=5, random_state=42); within-fold validation random_state=42+fold_idx.
- Locked checkpoint: models/checkpoints_constrained/checkpoint_constrained.pt = fold_0_best.pt, epoch 99, constraint_weight=0.1, selection = minimum fold-0 validation total loss (0.0241).
- Matched unconstrained: models/checkpoints_unconstrained/checkpoint_unconstrained.pt from checkpoints_5fold/fold_0_best.pt.
- Variant: constrained_mlp; 2,648,777 trainable parameters; ChemBERTa embeddings frozen .npy files.
- Preprocessing: FIR 0.5–40 Hz, training notch 50 Hz, FastICA with empty default exclusion, 4 s epochs at 90% overlap, per-subject z-score after epoching, Welch n_per_seg=256 at 500 Hz.
- Feature map: 2185-D, 0-based slices PSD [0, 380), bands [380, 475), interleaved connectivity [475, 2185); L_{conn} window [475, 1330).
- Dummy-batch inference (not clinical real-time or end-to-end latency): 0.602 ms/subject; dummy throughput 1660 subjects/s; peak dummy-forward GPU memory 50.2 MiB; checkpoint 31,876,749 bytes. Training wall-clock for the locked run is not logged.

- Expected locked direction outputs (fold-0 seed-42): $r = 0.908/0.851/0.918$ and 10/10 signs on TUH / OSF / P-ADIC against the training constrained signature.
- Master roll-up: `models/validation/complete_validation_report_v3.json`.

Appendix A.2. Model configuration

- Variant: `constrained_mlp` in `src/models/config.py` (`MODEL_VARIANTS['constrained_mlp']`).
- ChemBERTa input dim: `MODEL_CONFIG['drug_encoder']['input_dim'] = 384`.
- CVAE latent dim: `MODEL_CONFIG['cvae']['latent_dim'] = 128`.
- Hidden widths: EEG encoder [256, 512], drug encoder [256, 128], CVAE [512, 256]; `use_attention: False`.
- KL weight: `TRAINING_CONFIG['beta_kl'] = 0.01`.
- Constraint schedule: `warmup_epochs = 5`, `paper_constraint_weight = 0.1` in `run_constrained_training.py`. Implemented $\alpha_c(t) = \alpha_c^* \min(1, (t - T_w)/5)$ for $t \geq T_w$ with 0-indexed t , so $\alpha_c(T_w) = 0$.
- Parameter count: 2,648,777 (`models/validation/computational_requirements.json`).
- Fold-0 loss histories: `models/evaluation/constrained_evaluation_report.json` $\rightarrow$ `constraint_curves.fold_0` (Fig. A.1).

Appendix A.3. Pharmacodynamic priors and loss helpers

- Priors store: `src/drugs/pharmacological_embedding.py` (`DRUG_PHARMACOLOGY[*]['eeg_effects']`).
- Prior loader: `get_eeg_effect_prior(drug_name)`.
- Band constraint: `compute_band_direction_constraint` in `src/models/losses.py`.
- Connectivity constraint: `compute_connectivity_constraint`.
- Band / connectivity index slices used by external direction scoring: `BAND_SLICES` in `src/tuh/tuh_validation.py` (`delta` [380:399], `theta` [399:418], `alpha` [418:437], `beta` [437:456], `connectivity` [475:1330]).
- Flatten order (`api/utils/feature_processor.py` `flatten_features`): PSD [0:380], band powers [380:475], then for each of five bands coherence upper-triangle (171) then PLV (171), so full connectivity is interleaved on [475:2185]. Coherence and PLV are **not** contiguous pure blocks. The training connectivity constraint and direction-scoring window [475:1330] match `CONNECTIVITY_INDICES / BAND_SLICES['connectivity']`; that window mixes interleaved coh/PLV from early bands and must not be cited as an all-coherence block. Do not use `feature_block_ranges()['coherence_block']` as a pure-coherence map (it conflicts with interleaving).

Appendix A.4. Training data paths and checkpoint aliases

- Training arrays: `data/eeg_features/{AD,HC}*.npy` (PSD shapes (37, 19, 20) and (29, 19, 20)).
- Raw EEGLAB discovery: `EEGLoader.find_eeg_files` under `data/raw_eeg/{AD,HC}/`.
- Subject-level CV helper: `get_subject_level_kfold` (`n_splits = 5`, `random_seed = 42` from `DATA_CONFIG`).
- Constrained evaluation alias: `models/checkpoints_constrained/checkpoint_constrained.pt` (fold-0 best).
- Matched unconstrained alias: `models/checkpoints_unconstrained/checkpoint_unconstrained.pt` (from `checkpoints_5fold/fold_0_best.pt`).
- Simulation draws: `num_samples = 5` in `simulate_post_drug_eeg` for reported external runs.

Appendix A.5. External processing and validation artifacts

- TUH processing: `process_tuh_eeg.py` (`CROP_SECONDS = 320`, `TARGET_N_EPOCHS_FOR_PSD = 864`). For locked TUH results, the stored feature provenance in `data/tuh_features/processing_report.json` records `crop_seconds=60.0` and `apply_ica=false`; the 320-s value reflects the current script default and was not used to regenerate the locked TUH artifacts.
- CAUEEG processing: `src/tuh/process_caueeg_eeg.py`.
- Master report: `models/validation/complete_validation_report_v3.json`.
- Unconstrained battery: `models/validation/unconstrained_external_battery.json`.
- Prior-only baseline: `models/validation/prior_only_direction.json`.
- Fisher- z CIs for r : `models/validation/direction_r_fisher_ci.json`.
- α_c sweep: `models/validation/constraint_strength_sweep.json`.
- Latent probe: `models/validation/layer5e_caueeg_latent_probe_results.json`.
- Nested heads: `models/validation/layer5e_caueeg_classifier_head_results.json`.

Appendix A.6. Endpoint implementation notes

- Direction correlation key in JSON: `effect_magnitude_correlation`.
- Subject-bootstrap CIs: `models/validation/direction_r_subject_bootstrap_ci.json` (B=2000).
- Five-fold external direction: `models/validation/fold_external_direction_results.json`.
- Equal-weight fold ensemble: `models/validation/fold_ensemble/fold_ensemble_external_direction_results.json`.
- Ensemble subject bootstrap (B=2000, seed 42): `models/validation/fold_ensemble/bootstrap/ensemble_subject_bootstrap_B2000.json`.
- Prior vs ensemble incremental-value note: `models/validation/prior_vs_ensemble/` (paired Δr not supportable from stored prior-only artifacts).
- Domain-shift LR AUCs (2185-D): `models/validation/domain_shift/domain_shift_results.json`.
- Domain-shift forensics (feature families): `models/validation/domain_shift_forensics/domain_shift_forensic_summary.md`.
- Feature dimension map (interleaved connectivity): `models/validation/domain_shift_forensics/feature_dimension_mapping.md`.
- Raw 2185-D probe: `models/validation/layer5e_caueeg_raw2185_probe_results.json`.
- Magnitude score helper: `mean_drug_response`.
- Cohen's d helper: `_cohens_d` (sample variance, `ddof = 1`).
- Encoding probe scripts: `run_latent_probe_caueeg.py` (`PROBE_DISEASE_LABEL = 0`), `run_latent_classifier_head_caueeg.py`.
- Probe CV: `StratifiedKFold(n_splits=5, shuffle=True, random_state=42); N_PERM = 200`.
- Theta/alpha mean-fold AUC 0.675 vs pooled OOF AUC 0.673: `feature_probe_theta_alpha_dementia_vs_normal.cv.roc_auc.mean` vs `.cv.oof.roc_auc`.
- Raw 2185-D mean-fold AUC 0.699: `raw2185_logistic_dementia_vs_normal.cv.roc_auc.mean`.

Table A.1: Architecture of the locked `constrained_mlp` twin. Hidden blocks are Linear, BatchNorm1d, ReLU, Dropout ($p = 0.2$). Encoder/decoder output heads are linear (no output activation). Parameter count is from instantiating `UpgradedDigitalTwinModel` with `MODEL_VARIANTS['constrained_mlp']`.

Module	Input dim	Hidden widths	Output
EEG encoder	2185	256, 512	128
Drug encoder	384	256, 128	64
Fusion MLP	128+64+1	256	256
CVAE encoder	256	512, 256	$\mu, \log \sigma^2$ (128)
CVAE decoder	128+64+1	256, 512	2185
Total parameters	2,648,777 (all trainable)		

Table A.2: Measured computational quantities for the locked `constrained_mlp` checkpoint on an NVIDIA GeForce RTX 5060 Ti (16 GB). The 0.602 ms/subject figure is a dummy-batch inference measurement (batch of 16 random feature tensors; 50 timed forwards after 5 warmups) and is not a clinical real-time or end-to-end pipeline latency claim. Dummy throughput and peak GPU memory are from the same dummy-forward protocol. Training wall-clock for the locked fold-0 run is not stored in `constrained_evaluation_report.json`.

Quantity	Value
Trainable parameters	2,648,777
Checkpoint size	31,876,749 bytes
Dummy-batch inference	0.602 ms/subject
Dummy throughput	1660 subjects/s
Peak allocated GPU memory (dummy forward)	50.2 MiB
Python / PyTorch	3.10.0 / 2.10.0+cu128
NumPy / SciPy / scikit-learn	1.26.4 / 1.15.3 / 1.7.2
MNE / Transformers	1.11.0 / 5.0.0
Training wall-clock (locked fold-0)	not logged

Appendix A.7. Architecture, compute, and training curves

Layer widths and the measured parameter count (2,648,777) are listed in Table A.1. Dummy-batch inference time, checkpoint size, and package versions are listed in Table A.2. Fold-0 constrained train/validation total loss, MSE, KL, and unweighted $L_{\text{band}}/L_{\text{conn}}$ with $\alpha_c(t)$ are stored in `constraint_curves.fold_0`. Fig. A.1 plots those logged values.

Alternate-seed loss histories were not found in that report. The figure is a convergence diagnostic, not an external-performance claim.

Appendix A.8. Regeneration entry points

- `eeg_twin/Scripts/python.exe -m src.validation.run_prior_only_direction`
- `eeg_twin/Scripts/python.exe -m src.validation.compute_direction_r_cis`
- `eeg_twin/Scripts/python.exe -m src.validation.run_unconstrained_external_battery`
- `eeg_twin/Scripts/python.exe -m src.validation.run_alpha_c_sweep -eval-only`

Table A.3: Methodological safeguards and remaining scope boundaries. Rows for DeLong and chemistry conditioning summarize non-locked Phase C secondary artifacts; locked Layer-5* numbers are unchanged.

Issue	Safeguard in this study	What remains outside scope
Subject leakage	Subject-level stratified 5-fold; one subject in one test fold; per-subject z-score; probe scaler on training fold only	Features are extracted per subject before the split; no blanket no-leakage claim
Checkpoint selection	Minimum fold-0 internal validation total loss (epoch 99; val total 0.0241)	Historical preregistration is not independently proven
External assessment separation	External r , signs, and AUCs scored after freeze; not arguments to the saver	Chronological lock is documented, not audit-grade
Endpoint circularity	Concordance with the training constrained signature, not paired post-dose EEG	Empirical pharmacodynamic validation
Prior-only control	Constant literature vector: 10/10 signs at mean $r=0.636$. A non-locked fitted drug-conditioned linear imitation control also exists; targets are locked CVAE-simulated drug-minus-baseline effects	Imitation/control analysis, not independent biological validation
Continuous r instability	Five-fold means ≈ 0.37 – 0.39 ; non-42 r typically 0.08– 0.43 ; 10/10 is the sign endpoint	Locked 0.908/0.851/0.918 remain checkpoint-specific
Diagnostic baseline fairness	2185-D logistic 0.699 is a feature-space reference, not a CVAE score	No supervised deep 2185-D classifier
Multiplicity	Escalation sequence; no Bonferroni/FDR; secondary contrasts descriptive	Not a confirmatory closed testing family
DeLong	Paired DeLong on stored OOF scores ($n=727$); primary gaps latent/head vs θ/α and 2185-D	No multiplicity correction across pairs
Preprocessing prove-nance	FIR, notch, FastICA, 4 s / 90% overlap, per-subject z-score after epoching	Empty default ICA exclusion; native ds004504 reference not in code
Computational reproducibility	Table A.2 dummy-batch 0.602 ms/subject; 2,648,777 parameters	Locked-run training wall-clock not logged; dummy time is not clinical real-time
Chemistry generalization	Matched fold-0 ChemBERTa vs padded one-hot (non-locked); one-hot 10/10 signs	Not a large-library ChemBERTa claim; rematch r stays low

- `eeg_twin/Scripts/python.exe -m src.validation.run_delong_oof_probes`
- `eeg_twin/Scripts/python.exe -m src.validation.run_chemberta_onehot_ablation -eval-only`
- `eeg_twin/Scripts/python.exe src/figures/make_fig6_constraint_strength.py`
- `eeg_twin/Scripts/python.exe src/figures/make_graphical_abstract.py`

See also `paper/CODE_AVAILABILITY.md`.

Appendix A.9. Known missing artifacts

These items are not present in the locked record and are not invented here: locked-run training wall-clock; alternate-seed training curves; a public code URL and commit hash; the native EEG ref-

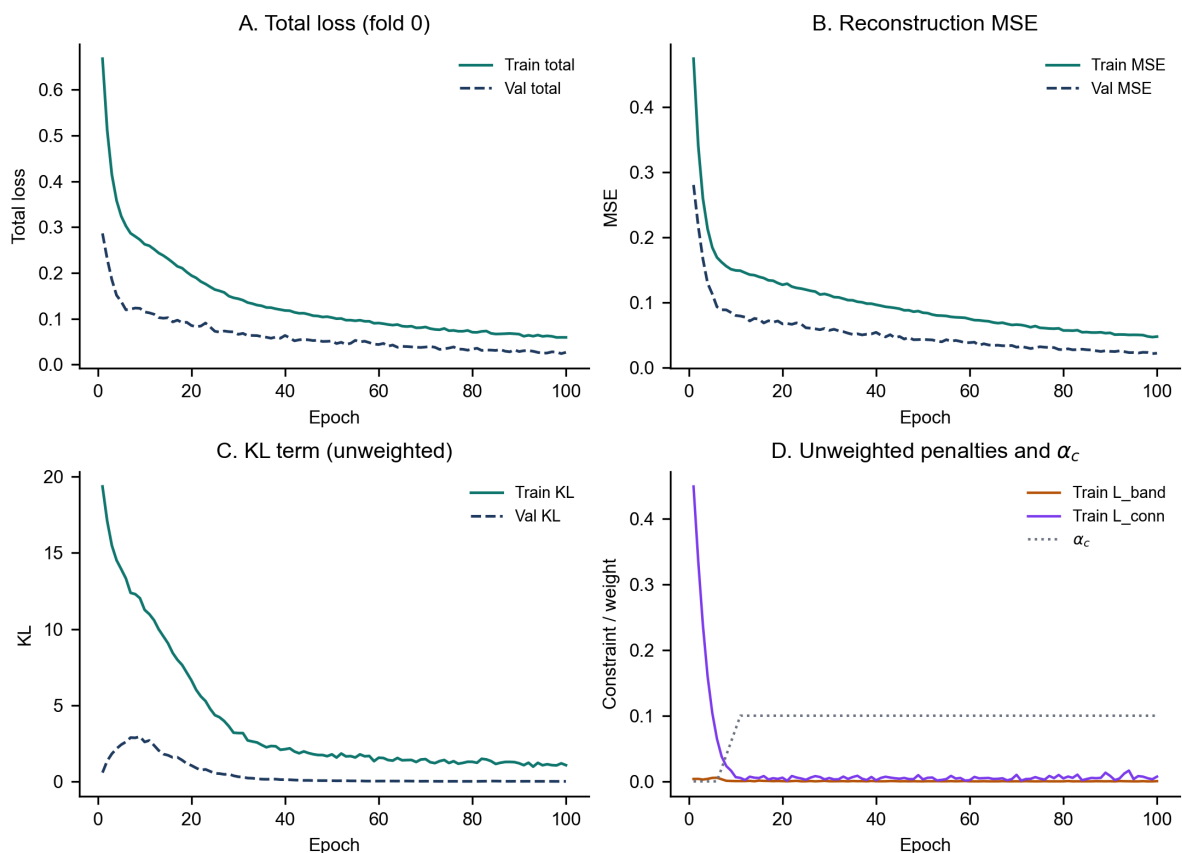


Figure A.1: Fold-0 constrained training curves from `constrained_evaluation_report.json` (`constraint_curves.fold_0`). (A) Train and validation total loss. (B) Reconstruction MSE. (C) Unweighted KL. (D) Unweighted L_{band} and L_{conn} with the implemented $\alpha_c(t)$ schedule. Validation total loss remains below training total loss on this fold (minimum validation total 0.0241 at epoch 99). Logged α_c is 0 through epoch 6 and 0.1 by epoch 11. Alternate-seed curves were not stored. Logged values only; not an external-generalization claim.

erence of the ds004504 files on disk; a supervised deep 2185-D classifier. Non-locked secondary Phase C artifacts now exist separately: DeLong on stored OOF scores (`models/validation/delong_oof_probe_results.json`) and a ChemBERTa-versus-padded-one-hot fold-0 ablation (`models/validation/chemberta_onehot_ablation/`). A non-locked fitted drug-conditioned linear imitation control is stored under `models/validation/linear_direction_baseline/` and is not a locked biological validation artifact. Table A.3 restates the corresponding scope boundaries.

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